LINUX SERVER ATTACK RESIDUE & FORENSIC RESPONSE SYSTEM

USE-CASE DOCUMENTATION

A report submitted in partial fulfillment of the requirements

for the degree of Bachelor of Technology in

Computer Science & Engineering / Cyber Security

Submitted by

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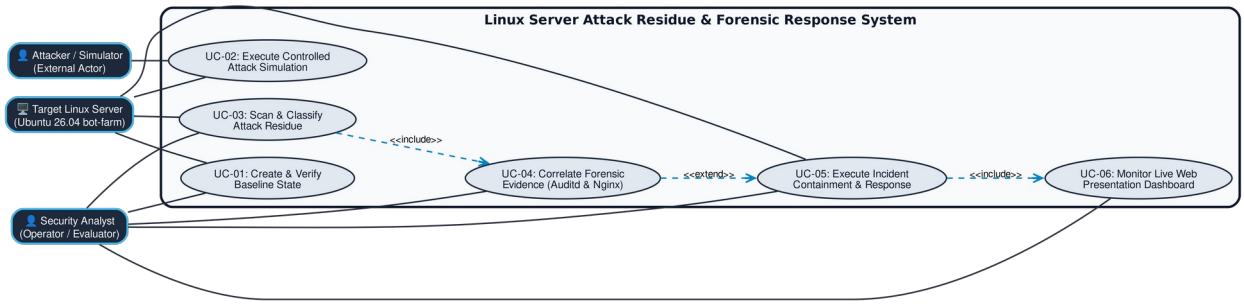
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1. Use-Case Diagram



2. Use-Case Descriptions

2.1 Example (filled) — Run Detection

Use Case ID	UC-01
Use Case Name	Record & Verify Baseline State
Actor(s)	Security Analyst (primary); Target Linux Server (secondary)
Description	Analyst records cryptographic SHA256 baseline state of monitored files, unit files, listening ports, users, and SSH keys.
Preconditions	Target host accessible; systemd and python 3.14 standard library operational.
Trigger	Analyst runs 'residue-system baseline create --name local' command.
Main Flow	1. Analyst initiates baseline recording. 2. System collects host state. 3. System computes SHA256 hashes. 4. Manifest stored at /var/lib/residue-system/artifacts/.
Alternate Flow	3a. Baseline already exists -> system verifies manifest cryptographic integrity.
Exception Flow	2a. System file unreadable due to permission -> logs warning and continues

	scan.
Postconditions	Trusted baseline JSON manifest saved and locked with 0600 permissions.
Priority	High

2.2 Blank table — copy one per use case

Use Case ID	UC-02
Use Case Name	Execute Controlled Compromise Simulation
Actor(s)	Attacker / Simulator (external); Target Linux Server (secondary)
Description	Simulator executes multi-stage attack (plants web shell, unauthorized update-service.service, and /opt/.cache/update binary).
Preconditions	Trusted baseline state recorded on target host.
Trigger	Analyst executes 'python3 attack/compromise_sim.py' command.
Main Flow	1. Simulator issues HTTP request to target. 2. Planted web shell drops executable. 3. Unit file registered. 4. Simulated backdoor active.
Alternate Flow	2a. Unit file already present -> overwrites unit file and updates state timestamp.
Exception Flow	1a. Target web server down -> simulation aborts with connection refused error.
Postconditions	Attack residues planted on target system.
Priority	High